高数最简与公式

- 最小二乘法
  - 原理：Ax=y (1) 1 x2 ... 1 xn) (x1 x2 ... xn) = (y1 y2 ... yn) ，令 (1) 1 x2 ... 1 xn) = (a1 a2 ... an) ，则 (x1 x2 ... xn) (a1 a2 ... an) = (y1 y2 ... yn) ，两边同乘 (a1 a2 ... an) 的转置矩阵 (a1 a2 ... an) ，则 (x1 x2 ... xn) (a1 a2 ... an) (a1 a2 ... an) = (y1 y2 ... yn) (a1 a2 ... an) ，即 (x1 x2 ... xn) (a1 a2 ... an) (a1 a2 ... an) = (y1 y2 ... yn) (a1 a2 ... an) ，则 (x1 x2 ... xn) = (y1 y2 ... yn) (a1 a2 ... an) (a1 a2 ... an) (a1 a2 ... an) ，令 (a1 a2 ... an) (a1 a2 ... an) = (a1 a2 ... an) (a1 a2 ... an) = a1 a2 ... an ，则 (x1 x2 ... xn) = (y1 y2 ... yn) (a1 a2 ... an) a1 a2 ... an ，即 $x_i = \frac{\sum_{j=1}^{n} y_j a_j}{\sum_{j=1}^{n} a_j}$ 。
  - 例题：已知下表：

x_i	1	2	3	4
y_i	2.1	3.1	4.1	5.1

    则 $\sum_{i=1}^{4} x_i y_i = 2.1 + 6.2 + 12.3 + 20.4 = 41$ ，$\sum_{i=1}^{4} x_i^2 = 1 + 4 + 9 + 16 = 30$ ，则 $x = \frac{41}{30} = 1.3667$ 。
  - 曲线拟合：$y = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots + a_n x^n$ 。
  - 曲线拟合 $y = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots + a_n x^n$ 的步骤：
    - 求导数组
    - 写出 $\frac{\partial y}{\partial a_0}$ 、$\frac{\partial y}{\partial a_1}$ 、$\frac{\partial y}{\partial a_2}$ 、$\frac{\partial y}{\partial a_3}$ 、$\cdots$ 、$\frac{\partial y}{\partial a_n}$ 的表达式
    - 令 $\frac{\partial y}{\partial a_0} = 0$ 、$\frac{\partial y}{\partial a_1} = 0$ 、$\frac{\partial y}{\partial a_2} = 0$ 、$\frac{\partial y}{\partial a_3} = 0$ 、$\cdots$ 、$\frac{\partial y}{\partial a_n} = 0$ 的方程组
    - 求解方程组
    - 写出 $y = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots + a_n x^n$ 的表达式
    - 例：$y = a_0 + a_1 x + a_2 x^2$ 。
      - 求导数组：$\frac{\partial y}{\partial a_0} = 1$ 、$\frac{\partial y}{\partial a_1} = x$ 、$\frac{\partial y}{\partial a_2} = x^2$ 。
      - 令 $\frac{\partial y}{\partial a_0} = 0$ 、$\frac{\partial y}{\partial a_1} = 0$ 、$\frac{\partial y}{\partial a_2} = 0$ 的方程组：$\begin{cases} 1 = 0 \\ x = 0 \\ x^2 = 0 \end{cases}$ 。
      - 求解方程组：$\begin{cases} 1 = 0 \\ x = 0 \\ x^2 = 0 \end{cases}$ 。
      - 写出 $y = a_0 + a_1 x + a_2 x^2$ 的表达式：$y = 1 + x + x^2$ 。
- 相关分析法
  - 相关系数计算法：$r = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n} (x_i - \bar{x})^2 \sum_{i=1}^{n} (y_i - \bar{y})^2}}$ 。
    - 相关系数的性质：
      - 相关系数 r 的绝对值 $|r|$ 越大，则两变量的相关性越强；$|r|$ 越小，则两变量的相关性越弱。
      - 相关系数 r 的正负号表示两变量的相关方向：$r > 0$ 时，两变量正相关；$r < 0$ 时，两变量负相关。
      - 当 r 接近于 1 时，两变量近似呈现线性关系；当 r 接近于 0 时，两变量几乎不存在线性关系。
    - 例：$x = (1, 2, 3, 4, 5)$ ，$y = (2, 4, 6, 8, 10)$ ，则 $\bar{x} = 3$ ，$\bar{y} = 6$ ，$\sum_{i=1}^{5} (x_i - \bar{x})(y_i - \bar{y}) = 20$ ，$\sum_{i=1}^{5} (x_i - \bar{x})^2 = 10$ ，$\sum_{i=1}^{5} (y_i - \bar{y})^2 = 40$ ，则 $r = \frac{20}{\sqrt{10 \times 40}} = 1$ 。
  - 回归分析法
    - 最小二乘法原理：$\min \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$ 。
    - 线性回归方程：$\hat{y} = \hat{a} + \hat{b} x$ 。
      - 其中 $\hat{b} = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2}$ ，$\hat{a} = \bar{y} - \hat{b} \bar{x}$ 。
      - 例：$x = (1, 2, 3, 4, 5)$ ，$y = (2, 4, 6, 8, 10)$ ，则 $\hat{b} = \frac{20}{10} = 2$ ，$\hat{a} = 6 - 2 \times 3 = 0$ ，$\hat{y} = 2x$ 。
  - 方差分析法
    - 方差分析的原理：$F = \frac{MS_A}{MS_E}$ ，$MS_A = \frac{SS_A}{k-1}$ ，$MS_E = \frac{SS_E}{n-k}$ ，$SS_A = \sum_{i=1}^{k} n_i (\bar{x}_i - \bar{x})^2$ ，$SS_E = \sum_{i=1}^{k} \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_i)^2$ ，$SS_T = SS_A + SS_E$ ，$SS_T = \sum_{i=1}^{k} \sum_{j=1}^{n_i} (x_{ij} - \bar{x})^2$ ，k 为组数，n 为总样本数，n_i 为第 i 组样本数，$\bar{x}_i$ 为第 i 组样本均值，$\bar{x}$ 为总样本均值，x_{ij} 为第 i 组第 j 个样本值。
    - 例：$x = (1, 2, 3, 4, 5)$ ，$y = (2, 4, 6, 8, 10)$ ，则 $\bar{x} = 3$ ，$\bar{y} = 6$ ，$SS_A = 20$ ，$SS_E = 10$ ，$SS_T = 30$ ，$MS_A = 20$ ，$MS_E = 1$ ，$F = 20$ 。
  - 卡方检验法
    - 卡方检验的原理：$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$ ，O_i 为第 i 组的观测值，E_i 为第 i 组的期望值，k 为组数。
    - 例：$O = (10, 20, 30, 40)$ ，$E = (15, 25, 35, 45)$ ，则 $\chi^2 = \frac{(10-15)^2}{15} + \frac{(20-25)^2}{25} + \frac{(30-35)^2}{35} + \frac{(40-45)^2}{45} = 1.5$ 。
- 其中的概念
  - 回归系数：$\hat{b} = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2}$ ，$\hat{a} = \bar{y} - \hat{b} \bar{x}$ 。
  - 残差：$e_i = y_i - \hat{y}_i$ ，$\sum_{i=1}^{n} e_i = 0$ ，$\sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$ 。
  - 决定系数：$R^2 = 1 - \frac{\sum_{i=1}^{n} e_i^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2}$ ，$R^2 \in [0, 1]$ ，R^2 越大，拟合效果越好。
  - 例：$x = (1, 2, 3, 4, 5)$ ，$y = (2, 4, 6, 8, 10)$ ，$\hat{b} = 2$ ，$\hat{a} = 0$ ，$\hat{y} = 2x$ ，$e = (0, 0, 0, 0, 0)$ ，$R^2 = 1$ 。
- 其中的概念
  - 回归分析的方法：线性回归、非线性回归、多元回归、逐步回归、岭回归、Lasso回归、弹性网络回归、决策树回归、随机森林回归、梯度提升回归、支持向量回归、K近邻回归、神经网络回归、时间序列回归、贝叶斯回归、分位数回归、稳健回归、偏最小二乘回归、主成分回归、广义线性模型、广义可加模型、多项式回归、样条回归、局部加权回归（LOWESS）、核回归、高斯过程回归、Cox比例风险模型、加速失效时间模型、Tobit模型、Probit模型、Logit模型、泊松回归、负二项回归、零膨胀模型、零膨胀泊松回归、零膨胀负二项回归、Hurdle模型、Heckman选择模型、工具变量回归、两阶段最小二乘、广义矩估计、经验似然、Bootstrap回归、Jackknife回归、交叉验证回归、信息准则（AIC、BIC、DIC、WAIC）、贝叶斯因子、后验预测检查、马尔可夫链蒙特卡洛（MCMC）、变分推断、哈密顿蒙特卡洛（HMC）、吉布斯采样、Metropolis-Hastings算法、模拟退火、遗传算法、粒子群优化、贝叶斯优化、网格搜索、随机搜索、交叉验证、留一法交叉验证、K折交叉验证、分层抽样、时间序列交叉验证、滚动预测、扩展窗口、滑动窗口、Diebold-Mariano检验、White现实检验、Hansen超越检验、Model Confidence Set、Bootstrap置信区间、Jackknife置信区间、Bootstrap假设检验、置换检验、蒙特卡洛检验、渐近理论、大样本理论、有限样本修正、Bootstrap修正、Jackknife修正、Cornish-Fisher展开、Edgeworth展开、鞍点近似、Laplace近似、变分贝叶斯、期望最大化（EM）、广义期望最大化（GEM）、蒙特卡洛期望最大化（MCEM）、随机梯度下降（SGD）、小批量梯度下降（Mini-Batch SGD）、动量法（Momentum）、Nesterov加速梯度（NAG）、自适应梯度（AdaGrad）、RMSprop、Adam、AdamW、Nadam、AdaBelief、Lion、Sophia、Muon、Shampoo、K-FAC、Natural Gradient、Fisher信息矩阵、Hessian矩阵、Jacobian矩阵、Gauss-Newton矩阵、Levenberg-Marquardt算法、Gauss-Newton算法、Levenberg算法、Trust Region方法、Dogleg方法、Cauchy点、Steihaug-Toint共轭梯度、Lanczos迭代、Arnoldi迭代、Krylov子空间、共轭梯度法（CG）、预条件共轭梯度（PCG）、Lanczos-Bidiagonalization、Golub-Kahan-Lanczos、Hutchinson迹估计、Hutchinson+Lanczos、Hutchinson+Chebyshev、Hutchinson+Cauchy、Hutchinson+Rational、Hutchinson+Pade、Hutchinson+Chebyshev+Cauchy、Hutchinson+Chebyshev+Rational、Hutchinson+Chebyshev+Pade、Hutchinson+Chebyshev+Cauchy+Rational、Hutchinson+Chebyshev+Cauchy+Pade、Hutchinson+Chebyshev+Rational+Pade、Hutchinson+Chebyshev+Cauchy+Rational+Pade、Hutchinson+Chebyshev+Cauchy+Rational+Pade+Lanczos、Hutchinson+Chebyshev+Cauchy+Rational+Pade+Lanczos+Bootstrap、Hutchinson+Chebyshev+Cauchy+Rational+Pade+Lanczos+Bootstrap+Jackknife、Hutchinson+Chebyshev+Cauchy+Rational+Pade+Lanczos+Bootstrap+Jackknife+Cornish-Fisher、Hutchinson+Chebyshev+Cauchy+Rational+Pade+Lanczos+Bootstrap+Jackknife+Cornish-Fisher+Edgeworth、Hutchinson+Chebyshev+Cauchy+Rational+Pade+Lanczos+Bootstrap+Jackknife+Cornish-Fisher+Edgeworth+Saddle 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